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Nano-Materials for Sustainable Energy (ChE 494)

Quiz 3: Principles of Solar Energy Harvesting

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40 Minutes

Total: 100

1. Sunlight is a form of "electromagnetic radiation" and the visible light that we see is a small subset of the electromagnetic spectrum.

(A) True

(B) False

2. Light is a type of quantum-mechanical particle called a/an .....

(A) Electron

(B) Photon

(C) Phonon

(D) Exciton

3. What is the relationship between the energy of a photon (E) and the wavelength of the light (λ)?

2.5 Inverse relationship

4. What is photon flux?

Defined as the number of photons per second per unit area

5. The Air Mass is the path length which light takes through the atmosphere normalized to the shortest possible path length (that is, when the sun is directly overhead).

(A) True

(B) False

6. Semiconductors act as conductors at low temperatures and insulators at higher temperatures.

(A) True

(B) False



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7. The band gap of a semiconductor is the minimum energy required to excite an electron that is stuck in its bound state into a free state where it can participate in conduction.

(A) True

(B) False

8. Doping creates N-type material when semiconductor materials from group IV are doped with ..... atoms.

(A) Group III

(B) Group IV

(C) Group V

(D) Group VI

9. What is law of mass action?

The product of the majority and minority concentration is a constant at equilibrium

$$n_0 p_0 = n_i^2$$

$n_i$  = intrinsic carrier concentration

$n_0$  = electron concentration

$p_0$  = hole concentration

10. When the energy of a photon is equal to or greater than the band gap of the material, the photon is absorbed by the material and excites an electron into the conduction band.

(A) True

(B) False

11. What is radiative (band-to-band) recombination?

An electron from the conduction band directly combines with a hole in the valence band and release a photon.

12. Auger Recombination involves ..... carriers.

(A) one

(B) three

(C) two

(D) four



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13. Auger recombination is most important at high carrier concentrations caused by heavy doping or high-level injection under concentrated sunlight.

(A) True

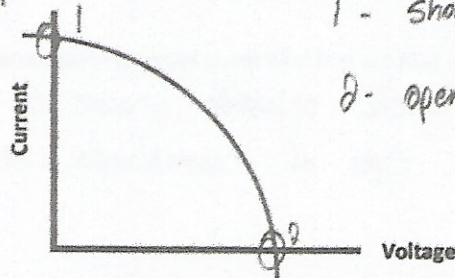
(B) False

14. Write the current-voltage (I-V) relationship for a P-N junction diode.

$$I = I_0 \left( e^{\frac{qV}{kT}} - 1 \right) \text{ for ideal diodes}$$

$$I = I_0 \left( e^{\frac{qV}{nkT}} - 1 \right) \text{ for non-ideal diodes}$$

15. The I-V curve of a solar cell is shown below. Indicate the short-circuit current density and open-circuit voltage?



1 - short circuit

2 - open-circuit voltage

16. Efficiency is defined as the ratio of energy output from the solar cell to input energy from the sun. Write the efficiency of a solar cell in terms of short-circuit current density, open-circuit voltage, fill factor, and input solar energy?

$V_{oc}$  = open circuit voltage

$I_{sc}$  = short circuit current

FF = fill factor

$\eta$  = efficiency

$$\eta = \frac{V_{oc} I_{sc} FF}{P_{in}}$$

17. What is the difference between direct and indirect bandgap semiconductors?

For direct bandgap, the conduction band and top of valence band lie at the same value of  $k$ . Indirect bandgap has different values of  $k$  for conduction and top of valence.



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18. Fill-out the following table for N-type and P-type silicon semiconductor?

|                   | N-Type Silicon | P-Type Silicon |
|-------------------|----------------|----------------|
| Dopant            | Group V        | Group III      |
| Majority Carriers | Electrons      | Holes          |
| Minority Carriers | Holes          | Electrons      |

19. What is a recombination process and how many types of recombination process are there?

Recombination is the opposite process to generation.

There are 3 types of recombination processes.

3

20. The absorption coefficient determines how far into a material light of a particular wavelength can penetrate before it is absorbed?

(A) True

(B) False